Symbols and Nomenclature

Geometric Constraints on Transition Dynamics in Recursive Language Systems

This reference sheet defines the principal symbols, geometric quantities, observables, and outcome classes used across the manuscript and supplement.

Symbol / term	Meaning
r	Primary control parameter indexing one axis of the probe state space.
α	Secondary control parameter indexing the other axis of the probe state space.
Fisher information metric	Riemannian metric used to define local geometry and distance between states.
Phase boundary	Estimated geometric separator between stable and unstable regimes.
Distance to phase boundary	Scalar distance from a state to the estimated phase boundary; primary organizing coordinate.
Local curvature / log curvature	Curvature proxy computed on the manifold; used as a local sensitivity or stress measure.
Transition ridge	One-dimensional profile or locus where transition probability is maximized near the phase boundary.
Lazarus score	Scalar observable summarizing trajectory-level compression; higher values indicate stronger trajectory-level compression.
Transition probability (k-step)	Probability that a trajectory changes regime within a fixed horizon k.
Approach direction to boundary	Signed early slope of distance-to-boundary along a trajectory; negative values indicate approach.

Outcome classes

Outcome	Operational definition
in_basin	Trajectory remains in its regime without boundary crossing or Lazarus activation.
seam_graze	Trajectory approaches the phase boundary within a small threshold but does not cross it.
seam_cross	Trajectory crosses the phase boundary without satisfying collapse criteria.
lazarus	Trajectory enters a high-compression or recovery-like regime without full collapse.
collapse	Trajectory crosses the phase boundary and exhibits at least one phase flip.
recovered	Trajectory crosses and later returns to the original signed phase.

Interpretive shorthand used in the manuscript

Distance gates, curvature modulates. Boundary proximity determines whether transitions are accessible; curvature shapes local transition likelihood within that accessible region.

Selection regime. Region of parameter space and initial-condition class associated with a dominant outcome behavior.

Projection of geometry. An observable such as Lazarus is interpreted as a lower-dimensional summary arising from the underlying Fisher-geometric structure.